PORTFOLIO · H1 FY2026

Gurinder Singh

From iOS Sandbox DRI to cross-BU AI-platform infrastructure — Vireo telemetry, the Darwin evolution framework, and the MAIA × Venmo track now folded into PayPal's Migration & Upgrade Program.

Role	Staff Software Engineer
Team	Pay With Venmo / Bazaar / DTVZ
Period	December 2025 – June 2026
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At a glance

Over the last seven months I evolved from being primarily the iOS Sandbox DRI on Pay With Venmo into a builder of cross-BU AI platform infrastructure — Vireo telemetry, the Darwin evolution framework, and the MAIA × Venmo track that's now folded into PayPal's Migration & Upgrade Program. The portfolio breaks into five arcs, each with executive sponsorship and merging downstream:

ARC	OUTCOME
Foundation — Mac-in-a-Mac CI/CD	First Mac-in-a-Mac at PayPal. **3.0× faster** iOS UI tests (PayWithVenmo 22m 38s → 7m 23s). Same VM substrate that now hosts Livery and Agent-0.
Universal Capture — claude-cosmos proxy	Anthropic-shape proxy routing to Cosmos with **sanitization + PII scrubbing + telemetry baked into the proxy**. The capture surface Vireo can't reach — hosted, ephemeral, third-party. Powers Osaurus, Hermes Agent, Agent-0, Cosmos-IDE, Devmo webapp, MAIA's network-layer trace.
Telemetry & Memory — Vireo	Owner of Devmo Workstreams 7 (Telemetry) + 8 (Memory). Multi-surface event capture (MCP/CLI/hooks/skills), n8n distillation, 33-field schema, hourly memory store. Cited by name in the Devmo HLD and Beta Workstreams.
Evolution — Darwin & ai-toolkit-registry-evolved	Authored the evolution framework (Seed → Zephyr → Polaris → Compass → Bloom → Fossil → Garden). Shipped **71 PRs across 2 milestones; 1,020 candidates, 19 judges, 678 eval examples** in a graph DB with lineage. AI Labs onboard.
Productionization — MAIA × Venmo	Drove Venmo track from cold-start to a Senior-VP-attended live demo that converted MAIA from "separate track" to the **unified PayPal Migration & Upgrade Program**. Authored the discovery layer (332 → 9,513 scored merchants in one weekend) and the injection script that took MAIA from fixtures to real merchants.

Plus shipped joy and polish: **DX Sounds MCP server** (merged into the PayPal AI Hub), the **"How to Have the Most Fun with AI"** talk that launched **FunAI Hack Hour** as a recurring forum, the **Prompt Enhancer with zero-trust guardrails** in Devmo Beta, the **OmniGraph / PWV Knowledge Graph** three-track framework, and the **DEVMO DOSSIER** competitive-intel field report across PayPal+Venmo Slack. Plus **Mercury 2025 / Hackathon-2025** — the autonomous-merchant blueprint that turned out to be the imagination doc for everything in H1 FY2026.

Mid-year update (through June 2026): the six weeks since the May checkpoint turned momentum into shipped infrastructure — **Darwin productionized** (callable from any repo, CI-gated, first end-to-end "Bloom" PR), **Vireo v2** (RAG + per-sub-agent telemetry + OpenInference/OTel export + cross-language memory), the **MAIA × Venmo track folded into the single Migration & Upgrade Program** and now moving to live cohort pilots, a new shared `video-toolkit` repo, and a **security-review exchange** that put the agent-evolution work in front of the enterprise's AI-tools evaluation. See Since the May checkpoint.

By the numbers

METRIC	VALUE
iOS UI test speedup proven	**3.0×** (22m 38s → 7m 23s on PayWithVenmo plan)
Merchant corpus expansion	**332** → **9,513** scored US merchants in one weekend
Prime Venmo V1 candidates identified	**476** ("bullseye": simple checkout + PayPal already integrated + US + web)
Independent harvest campaigns built	**18** (CommonCrawl + AWS Athena + specialty hunters)
Darwin evolution PRs shipped	**71** across m1 (37) + m2 (34)
Darwin fossil-record graph DB	**1,020 candidates** · **19 judges** · **678 eval examples** with full lineage
MAIA executive-demo attendees	**14** (required + optional senior leaders)
MAIA live demo runtime	**2 parallel merchant sites, V5 → V6 + Venmo, ~20 minutes on stage**
PayPal AI Hub MCP servers I authored	**1** (DX Sounds — 8 themes, 14 tools)
PayPal-Venmo repos I authored or co-author	**8** (Vireo, vireo-telemetry, developer-experience-sounds, apple-virtualization-ci-cd, claude-cosmos, cosmos-code-mcp, video-toolkit, Darwin)
Personal `gurisingh_paypal` repos	**5+** (darwin, ai-toolkit-registry-evolved, paypal-agent-zero, devmo-beta-service, Cosmos-IDE)
Cross-BU Slack channels actively driving conversation in	**#v-devmo, #smb-fs-ai, #ai-labs, #v-teen-ios, #ai-accelerator, #v-merchant-sandbox, #tmp-maia-venmo-xfn, #pwv-bazaar**
MAIA run captured end-to-end via Vireo v2.1 (May 15)	**53 events** · **45 traces** · **40 tool calls** · **10 skill events** · **100% success in ~8 min**
Darwin invocation modes for any repo (no fork required)	**3** (reusable workflow · composite action · `repository_dispatch`)
Darwin real-repo evolution runs (mid-year)	~**120** against the Pay-with-Venmo skill registry (`skills-by-pwv`)
Devmo Prompt-Enhancer guardrail test battery	~**90 tests** including a prompt-injection cohort

Scope expansion (the calibration framing)

Where I started December 2025: Staff Software Engineer on Pay With Venmo, Bazaar/DTVZ. Operational focus: iOS Sandbox DRI, TestFlight pipeline, Ladybug debug menu, daily E2E call ownership, release manager duties for PWV Sandbox.

Where I am June 2026: Same title, materially expanded scope. I'm now:

- **Owner** of the Vireo telemetry + memory layer — now **v2** (RAG, per-sub-agent telemetry, an OpenInference/OpenTelemetry export, and a cross-language Python + Swift memory store) — used across Devmo (Venmo) and instrumenting live MAIA runs (PayPal).
- **Originator** of the Mac-in-a-Mac substrate that Livery and Agent-0 run on top of — and the sanctioned, VM-isolated containment that let me run external-agent (Hermes) research safely, as CDC/InfoSec confirmed in June.
- **Architect** of Darwin, now **productionized**: invokable from any repo three ways (no fork), CI-gated with a score-delta PR comment, first clean end-to-end "Bloom" suggestion PR landed, ~120 real-repo evolution runs, and a home in the PayPal-Venmo org. The AI Toolkit lead, AI Labs, and Devmo are converging on it.
- **Named to the MAIA "build" core** (a Singapore MAIA architect / Gurinder Singh / a Sr SWE on my team). The Venmo track is now folded into *"the one MAIA roadmap"* (the MAIA PM, Jun 24) inside the single PayPal Migration & Upgrade Program the program lead set up on **May 4 2026** — and we're moving from demo to **live cohort pilots**.
- **Knowledge-transfer lead** — ramped a Sr SWE on my team as Venmo MAIA contributor (with Vireo write access) while continuing as iOS Sandbox DRI; Mac-in-a-Mac continuing toward a peer engineer + the iOS sponsor's team.

This isn't title inflation talk — it's an org-chart catch-up note. The work crosses **3 BUs** (Venmo, SMB-FS, LE/GSA), is sponsored at the **Senior Director** / **VP-adjacent level**, and the post-demo follow-ons are *consequences* of my output, not just my todo list.

The five arcs

1. Foundation — Mac-in-a-Mac CI/CD (`PayPal-Venmo/apple-virtualization-ci-cd`)

I was **first at PayPal to run a macOS guest VM inside a macOS host** using Apple's Virtualization.framework. The original target was the #1 iOS CI bottleneck: long UI test runs. The first measured result was **PayWithVenmo plan 22m 38s → 7m 23s (3.0× speedup)** on existing hardware — no cloud, no new minis, no test modifications. Three proof runs with full xcresult bundles, screenshots, process lists, and timing data lived in `docs/` from day one.

What it became:

- **CI substrate:** every iOS team can shard test plans across host + 1–2 child VMs per Mac mini. 120-test suites become 3 shards of 40 running in parallel.
- **Agent runtime:** dedicated macOS VMs give Agent-0 hardware acceleration, macOS-use control, and "always-on" mode — the alternative to Docker/Rancher that mirrors a real developer's toolset.

- **Livery's runtime primitive:** I pitched the Livery lead directly on Livery running macOS-VM automation runs + Livery running MAIA on **April 9 2026**. That's now the convergence pattern.
- **Cross-team adoption:** a peer engineer picked up the lume integration and extended it; a reviewer + a reviewer reviewed; a senior engineering director brokered; an iOS leadership sponsor signed on for sponsorship on **May 5** with *"i love it; i am sold on the performance."*

Confluence HLD: *Draft HLD — Parallel iOS UI Test Execution Using Apple Virtualization.framework* (VAC, page `2860853837`). **Repo branch:** `feature/parallel-ui-tests` with `GUIDE.md`. **Status:** in active handoff. Operational ownership moving to the iOS leadership team and a peer engineer; my fingerprints are now architectural.

2. Universal Capture — claude-cosmos proxy (`PayPal-Venmo/claude-cosmos`)

> ***The capture surface where Vireo can't reach.***

What it actually is. A Claude-flavored proxy that routes any Anthropic-shaped request to PayPal's Cosmos LLM endpoints, with **sanitization, PII scrubbing, and telemetry built into the proxy itself**. The thing it unlocks isn't just "Claude on Cosmos models" — it's **"any agent, anywhere, instrumented and safe."**

Where it's used today.

- **Osaurus** (Mac Studio local inference)
- **NousResearch Hermes Agent** (self-evolving)
- **Agent-0** (always-on autonomous loop)
- **Cosmos-IDE** (`gurisingh_paypal/Cosmos-IDE`)
- **Devmo agent webapp** — any LLM call out-of-band of a hook or MCP surface
- **MAIA's network-layer telemetry path** (the Proxy/Network option in `DTVZ-506`)

Why it's the cleanest substrate.

LAYER	CAPTURE	LIMITATIONS	CLAUDE-COSMOS ANSWER
CLI / IDE	Vireo (hooks, MCP, skill instructions)	Only catches local runs that touch hooks/MCP	Out of scope — Vireo wins
Hosted / ephemeral / webapp	Datadog / Langtrace / vendor SDKs	Expensive at scale, no PII story baked in	**claude-cosmos** does it for free; sanitization + PII scrub + telemetry in-proxy
Untrusted / third-party agents	None — black box	Can't observe what doesn't reach you	**claude-cosmos** — every call has to come through it to reach Cosmos at all

The full picture. Vireo catches what an agent *intends* to do on local CLI/IDE; claude-cosmos catches what *every* agent actually *says to the model* over the wire. Together: intent + ground truth across local, hosted, ephemeral, and untrusted environments. **There's nowhere an LLM call can hide.**

Jira: `DTVZ-331`. Companion: `cosmos-code-mcp` (`DTVZ-330`) for the MCP-protocol surface.

3. Telemetry & Memory — Vireo

What it is. Agent-agnostic event capture for AI dev workflows. Four capture surfaces (MCP server, CLI `vireo-send`, shell hooks, skill instructions). 33-field schema with 4 required fields (`intent`, `outcome`, `repo`, `tokens_out`). n8n ingestion → events table → daily GitHub archive. Hourly n8n workflow distills raw events into session summaries, repeated patterns, and learning candidates across five memory kinds (procedural, failure, preference, semantic, episodic).

Where it lives.

- Devmo Workstreams 7 (Telemetry) + 8 (Memory) — Confluence page `2900859692`
- Repos: `PayPal-Venmo/Vireo`, `PayPal-Venmo/vireo-telemetry`
- Jira epic: `DTPLTM-8222` — *Devmo - Telemetry (Vireo)*

Adoption. A colleague, an SMB-FS colleague ("this looks fantastic"), a teammate and the PWV-Bazaar PR/CR-party team have already onboarded via `./start.sh`. The MAIA program is now using Vireo for its post-demo "Exhaustive Tracing for Claude Code Session" telemetry track (`DTVZ-506`) — Vireo is the substrate the unified migration program is being instrumented on.

4. Evolution — Darwin (`gurisingh_paypal/darwin`) + ai-toolkit-registry-evolved

Thesis. Anything with a prompt in, an output out, and a judge that can score it is GEPA-evolvable. Darwin is the home for that loop: **Evolve → Evaluate → Preserve → Explain → Open PR.** Regressions are not failure; they are scientific signal.

Vocabulary I introduced and that has stuck:

- **Seed** — the starting artifact (skill, prompt, PRD, design, workflow)
- **Zephyr** — the evolutionary run (wave of mutation candidates)
- **Polaris** — the North Star objective
- **Compass** — the eval/judging criteria
- **Bloom** — the winning candidate under the current compass
- **Fossil** — a preserved non-winner; recorded evidence for future runs
- **Garden** — the full ecosystem of artifacts, runs, history, lineage
- **Gardener** — Darwin + the humans/agents tending the system

Concrete output as of May 8: 71 PRs across two milestones on `ai-toolkit-registry-evolved` (m1 = 37, m2 = 34, plus a `regressed` label cohort). Every PR ships with a narrated video, score table, candidate arena (all fossils + the bloom), judge prompt, eval dataset, and full diff. The fossil record — every candidate that ever competed across both milestones — is stored in a graph DB in the Darwin repo with full-text search: **1,020 candidates · 19 judges · 678 eval examples · cross-milestone lineage tracking.**

Live POC surfaces:

- Visual perspective: `web-three-beta-19.vercel.app`
- Wrapped via internal n8n: `aiplatform.dev51.cbf.dev.paypalinc.com/n8n/webhook/darwin-vercel`

Doctrinal alignment with NousResearch's `hermes-agent-self-evolution` and Imbue's Darwinian Evolver — Darwin is the PayPal-flavored implementation of the same evolutionary-loop idea, with our own Compass/Bloom/Fossil vocabulary and our own GEPA + DSPy under the hood. The AI Toolkit lead is the scope partner; AI Labs is the next adoption frontier.

5. Productionization — MAIA × Venmo

The arc that connected everything. From cold-start to executive demo to unified program in under three weeks.

Timeline.

DATE	BEAT
Apr 10	Reverse-engineered MAIA's 4-step pipeline (`aim_init_check → aim_convert_router → aim_correction → aim_paypal_ft`). Shipped initial 320-merchant test bed sourced from CommonCrawl + BuiltWith across 17 platforms and 19 verticals.
Apr 13	Architecture sync with the Singapore MAIA lead, a Venmo product leader, the AI Toolkit lead, my manager. Decisions locked: opt-in Venmo, reuse SPB eligibility endpoint, web-only V1, Claude Sonnet recommended.
Apr 14	**Weekend crawl story: 332 → 9,513 scored US merchants.** 18 independent harvest campaigns (CommonCrawl + Athena + specialty hunters: `enterprise_hunter` 686 leads / 96% S-Tier, `legacy_hunter` 746 / 98% exclusive, `mega_r2` 4,016 cross-validation). **476 prime Venmo V1 candidates in the bullseye.** Four interactive D3.js knowledge graphs (Gateway Ecosystem, Merchant Intelligence Overview, Venmo Opportunity Map, Harvest Lineage) and a 1:45 narrated Remotion explainer video shipped same weekend.
Apr 17	First live MAIA pipeline E2E run.
Apr 21	AIM_TOKEN received from the Singapore MAIA lead. Demo invite accepted.
Apr 27	Live strategy call with **the Singapore MAIA architects**. Ran the Venmo functional test on the pizza-site fixture *plus* a real merchant via my injection script. Architecture interrogated; commitments extracted on dashboard access, EVAL repo timeline, skill extensibility, app-switch dedup.
Apr 29 (12:05–12:30 CT)	**THE DEMO** — `Eyes of Texas (11) MTR` , Austin. 14 required/optional senior leaders. Live: MAIA took a real V4/V5 merchant codebase, upgraded to JS SDK V6, and enabled Venmo — two parallel sites in ~20 minutes on stage.
Apr 30	The MAIA PM's recap email. Leadership in the room all +1.
May 4 (10:09 AM)	**The program lead unifies the program**: *"Venmo upgrade should be part of the migration and upgrade strategy that is in the works. It should not be a separate track."* Adds the GTM lead (GTM) + a partner.

DATE	BEAT
May 4	The MAIA PM: *"fully aligned, this should be one program, not a separate track. I'll setup next steps with Bo on the SMB side… and Karin on the trust-first UX."*
May 6	`DTVZ-506` opened — *MAIA Telemetry: Exhaustive Tracing for Claude Code Session* — direct follow-on from the demo.

My six unique contributions.

1. **Merchant discovery layer** — turned an inert 320-fixture sandbox into a 9,513-merchant scored corpus across 18 harvest campaigns. Singapore had zero real-world telemetry; I gave them ground truth.
2. **Injection script** — took MAIA from "works on our pizza-site fixture" to "works on real merchant markup we pulled from the wild." Reviewed and signed off by the Singapore MAIA architects on Apr 27.
3. **Four D3.js knowledge graphs + the 1:45 Remotion explainer** — turned a CSV into a leadership-grade artifact. This is the *visible* knowledge-graph asset for the program.
4. **Architecture interrogation** — the Apr 27 agenda I handed my manager is the document that extracted the load-bearing commitments from the Singapore MAIA lead.
5. **Vireo-into-MAIA wiring** — instrumented day one so post-demo telemetry data starts flowing. That's what produced `DTVZ-506`.
6. **Demo execution** — friction-bypass toggle, "Stay to the plan" hook reveal, live V5→V6→Venmo on two parallel sites in 20 minutes. The MAIA PM drove narrative, a Venmo product leader narrated screen, I ran the agent.

Receiving-end reactions in the room (verbatim).

> ***A Venmo leader*** *— "You're going to need to optimize for the developer believing they're in control… we both us and our partner could sleep well at night using this."* ***A Venmo product leader*** *— "Any merchant-initiated change should require them to essentially upgrade to the latest… and the other is also true. Anytime we upgrade, we want to attach Venmo."* ***A Venmo lead*** *— "This is all so awesome to see… ensuring the trust is going to be super important."* ***The MAIA PM's close*** *— "We ease anxiety and then we end with delight."*

Bonus — Mercury 2025 / Hackathon-2025: the imagination doc that turned into the roadmap

Repo: `github.paypal.com/gurisingh/Hackathon-2025` (15 commits, Jul–Aug 2025)

A hackathon-grade autonomous AI agent suite operating across the merchant lifecycle — predictive order fulfillment, multi-merchant discovery, agentic shopping (Kolors Virtual Try-On for fashion), fabric/image-gen pipelines, MCP-driven storefront orchestration. *"REST APIs are of the past. The future is agentic and smart AI."*

What it has become, in retrospect, is the **proof-of-concept blueprint** for everything that's followed in H1 FY2026:

- **MAIA** — the migration agent that *enables* this kind of autonomous merchant world
- **Vireo + claude-cosmos** — the telemetry to *understand* it
- **Darwin** — the loop to *improve* the skills running it
- **Mac-in-a-Mac** — the runtime substrate to *host* the always-on autonomous agents

Mercury 2025 was *literally just the beginning.* H1 FY2026 has been the foundational layer.

Since the May checkpoint (May–June 2026)

The May 11 portfolio closed at the moment MAIA was unified into the migration program. The six weeks since turned that momentum into **shipped infrastructure**, a **production-pilot motion**, and a **security-review exchange** that elevated the agent-evolution work to an enterprise audience. The five arcs above are the H1 foundation; this is the mid-year delta.

MAIA → one program, now moving to live pilots

DATE	BEAT
May 14	Scoped + closed the MAIA × Venmo P0 with my manager (`DTVZ-544` … `555`): toolkit credentials/infra, the **Cosmos LLM proxy for telemetry + payload visibility**, the Venmo eligibility approach with the AI Toolkit lead, the v6 SDK-wrapper review with a Singapore MAIA architect's team, the functional test via my injection script, the executive deck, and **a Sr SWE on my team's ramp-up** as Venmo MAIA contributor.
May 15	Instrumented a full MAIA run end-to-end through **Vireo v2.1 — 53 events, 45 traces, 40 tool calls, 10 skill events, 100% success in ~8 min** — with per-sub-agent telemetry (which agent ran, e.g. `paypal-venmo-migrator` / `paypal-migration-code-reviewer` , a privacy-safe title, success/error/escalated outcome, **no raw prompt or response text**). This is `DTVZ-506` "exhaustive tracing" landing.
Jun 8	The MAIA PM (back from leave) named me to the MAIA **build** core: *"build (a Singapore MAIA architect / Gurinder Singh / a Sr SWE on my team)."*
Jun 10	Opened the MAIA execution tickets (`DTVZ-670` … `675`): Cursor Venmo V6 integration merge, telemetry-analytics merge, a **v1 training-dataset export**, the first **merchant-pipeline batch with a Sr SWE on my team**, a merchant failure taxonomy, and the `maia_next500` queue.
Jun 17	MAIA "cohorting" working session — move from demo to a **first contained live experiment**: pick 1–2 low-risk cohorts, agree success criteria, name owners + dates.
Jun 23–24	"Sutter Pay With Venmo + MAIA" demo; the MAIA PM formalizes the consolidation — *"plug Venmo into the one MAIA roadmap you own"* — citing a mandate from the roadmap review with senior leadership and the follow-up demo (**a Venmo product leader + a Venmo leader behind it**).
Jun 26	The standalone Venmo working group is folded up — *"Canceling as we're looking to combine with other efforts."* Venmo now lives inside the single Migration & Upgrade Program, exactly as designed on May 4.

Darwin — productionized (the #1 May "what's next," delivered)

- **Invokable from any repo, three ways** — a reusable workflow (`darwin-evolve-callable.yml@v1`), a composite action (`PayPal-Venmo/Darwin@v1`), and a `repository_dispatch` listener: *"teams don't have to fork Darwin."*
- **PR-gate automation**: on any PR that touches a skill, Darwin auto-runs (GEPA-mature, now supporting organic / synthetic / mixed eval data via `data_mode` + `mix_ratio`) and posts a sticky comment with the **score delta vs. base** plus a link to a proposed enhancement PR — warn-only during the sandbox phase.
- **First clean end-to-end run** in `ai-toolkit-registry-evolved` : demo PR #45 → a Darwin-authored **"Bloom" suggestion PR #46** (PR detection → self-hosted runner → score → suggestion).
- **~120 evolution runs** against the real Pay-with-Venmo skill registry (`skills-by-pwv`) — a second, real-repo proof point beyond the sandbox.
- Darwin now lives in the **PayPal-Venmo org with CI enabled** (`DPEIR-4171`). My own framing in the June security review: *"the piece I'm proudest of… some of the most remarkable work I've had the chance to bring to the team."*

Vireo — v2 (RAG · sub-agent telemetry · OTel/OpenInference · cross-language memory)

- Built **RAG into Vireo**; shipped **v2.1** per-sub-agent telemetry (above).
- **Memory-stack v2** (Vireo PR #4): a cross-language **Python + Swift** hybrid retrieval + lifecycle system with a MemFS-style markdown export mirror and eval gates.
- Shipped a unified **OpenInference + Vireo V2 → OpenTelemetry** span-attribute conversion layer (Devmo PRs #128 / #165) — the OTel-native direction from the May plan, now in code, so Livery / Datadog / Langtrace can consume Vireo events.

Devmo — Prompt Enhancer + Guardrails, productionized into the canonical repo

- My two `devmo-beta-service` PRs migrated into the canonical **`PayPal-Venmo/devmo`** repo (PRs #59 Guard Rails, #60 Prompt Enhancer). The guardrails ship with **~90 tests**, including a prompt-injection battery.
- New scoped work: an **OpenAI-compatible Devmo endpoint for Xcode** (`DTVAI-18`), **prompt-enhancer skill auto-selection** (`DTVAI-19` / `21` , merged), and an **SSO login-visibility dashboard** (`DTVAI-20`).
- Raised a skill **supply-chain safety policy** after ~820 malicious skills were found in a public hub — proposed a "no URLs in skills" rule (with a Devmo colleague + the AI Toolkit lead).

`video-toolkit` — new shared repo (`PayPal-Venmo/video-toolkit`)

Open-sourced internally after the **May 12 team demo**: a **Remotion 4.0 + Google Veo** / **Chirp3-TTS** pipeline for explainer videos, podcasts, PR-demo clips, and Darwin evolution videos — runnable from GitHub Actions with no local setup. The per-PR narrated-video capability is now a tool anyone on the team can use.

Hermes security exchange — research turned into an enterprise input

InfoSec/CDC flagged my use of the open-source **Hermes** self-evolution agent (the lineage behind Darwin). Rather than just close the ticket, I posted a full voluntary accounting: **99.88% of traffic stayed on Cosmos or my internal proxy; the entire external footprint was 12 sessions / 37 messages (each enumerated); no MCP servers; one stray GitHub token already rotated** — and noted Hermes ran *inside an Apple VM, not on* *my host* (my own Mac-in-a-Mac substrate). I removed the agent and packaged a reusable `hermes-cleanup` skill.

> ***The CDC manager (CDC manager)*** *— "Your sharing is definitely beneficial and is worth having the broader conversation with [the] Security Consulting team who is performing an evaluation of AI tools across the enterprise."* ***A colleague*** *— "pushing AI culture to its limits!"*

Knowledge transfer — what's already handing off

This is the *handoff-in-progress* table — the thing that makes the scope expansion sustainable.

SURFACE	SUCCESSOR / NEW OWNER	STATUS
Venmo iOS Sandbox (DRI, TestFlight, Ladybug, daily E2E)	**A Sr SWE on my team** (Sr SWE, PWV Bazaar)	KT doc drafted (`DTVZ-339` closed Apr 15). Now also **ramped as Venmo MAIA contributor** (`DTVZ-555`) with **Vireo write access** and the MAIA demo flow (`LeadIntelligenceVault` testbed, `vireo_merchant_runner.py`); ran the first merchant-pipeline batch in June.
Mac-in-a-Mac / **Apple Virtualization CI/CD**	**The iOS leadership team** (formal sponsor) · **a peer engineer** (extended via lume) · **the Livery lead** (consumer for Livery)	Pitched May 5. The iOS sponsor committed: *"I can ask my team to take a look and give you some feedback."*
MAIA Venmo track telemetry	Migrating into the unified Migration & Upgrade Program	Vireo wired in via `DTVZ-506` (exhaustive tracing, sprints S340→S343); Vireo v2.1 captured a full run at 100% success (53 events / 45 traces / 40 tool calls).
PWV Knowledge Graph / OmniGraph	Three-track framework (Track 1 = Context Aggregation, peers, avoiding overlap)	Three-Track memo posted Mar 31, `DTVZ-335` .
Vireo telemetry consumers	Devmo platform team (the Devmo lead) · MAIA program	Self-serve via `./start.sh` . Multiple teams onboarded.

Team demo — delivered (May 12 2026)

Delivered the 45-min team session on **May 12 2026** — a TL;DR with MAIA plus actionable takeaways for day-to-day use, and the venue where I shared the new **`video-toolkit`** repo before open-sourcing it internally.

Goal: TL;DR with MAIA → actionable takeaways the team can use in their day-to-day.

Outline

1. **TL;DR via MAIA** (≈ 5 min) — Show the executive demo highlights. *"This is what we did, this is probably going to run throughout the year, this is how it works today, this is what's next."*
2. **Ephemeral Bootstrapping** (≈ 5 min) — How MAIA's per-session pattern works. Why ephemeral is a deliberate happy-medium between LLM autonomy and merchant transparency/ownership. The "Stay to the plan" hook.
3. **How these AI skills are built** (≈ 8 min)
    - Origin story: UI/UX of `CLAUDE.md` and `AGENTS.md`
    - Multi-sub-agent architecture
    - "Most of the logic lives in markdown text, no code — glimpse of what the future of coding looks like"
    - This is how to build something like this for your team
4. **Gotchas with Vineyard Vines** (≈ 5 min) — The feedback-loop and context-window edge cases. *"For VineyardVines, I spent an hour dealing with IFRAME chaos while using the baseline inject skill I shared with the team."* What I learned, what the team should anticipate.
5. **How you can use it in your day-to-day** (≈ 8 min)
    - **Graph RAG** — code/context graphs to give your agents memory
    - **Vireo** — local, offline, anonymous telemetry. `./start.sh` installs in 2 min.
    - The Soul Stack (`SOUL.md` , `STYLE.md` , `SKILL.md`) pattern
6. **Maybe some Remotion videos** (≈ 4 min) — Show the kind of artifacts you can produce per PR.
7. **Q&A** (≈ 15 min)

Bonus credit, courtesy of the room: *"Tested this like 30 times with my manager."*

What's next

For the second half (July–December 2026):

1. **MAIA live cohort pilots** — run the first contained cohorts (the Jun 17 plan), ship the merchant-visible automated test harness, land **Braintree support** in MAIA, and feed vendor metrics back into continuously-improved skills.

2. **Darwin adoption beyond the sandbox** — flip the PR gate from warn-only to enforcing on a first sponsored registry, onboard AI Labs, and add a second use case (agent personas, tool descriptions, slash commands — anything GEPA-eligible).
3. **Vireo OTel** / **OpenInference GA** — graduate the export so Livery / Datadog / Langtrace consume it in production, and harden the cross-language (Python + Swift) memory-stack v2.
4. **EMU + Harness CI migration (by June 30)** — get Vireo, Darwin, and the devmo repos migration-ready ahead of the CTO-wide deadline.
5. **Sandbox DRI continuity + KT** — close out the iOS Sandbox login / deep-link bugs and the Braintree app-to-app spikes, ship the sandbox telemetry instrumentation (`DTVP-4636`), and keep a Sr SWE on my team ramped on Venmo MAIA.

Asks:

- **Calibration conversation** — given the cross-BU scope, the MAIA build-core naming, and the *"one MAIA roadmap"* framing, I'd like a structured conversation about title/scope progression this cycle.
- **EMU GitHub access for Cosmos-IDE** / **Darwin** — the `gurisingh_paypal` namespace is where my evolution work lives; a cleaner collaborator path matters even more with the June 30 EMU migration in flight.
- **Service account for Gemini ADC** — still on my personal ADC for the Prompt Enhancer; would prefer a sanctioned path before this scales.

Appendix A — Project register

For everyone-facing reference. All items shipped or in-flight in this period.

Author / Owner

PROJECT	REPO / DOC	TYPE
Vireo (Telemetry)	`PayPal-Venmo/Vireo`	MCP server + CLI + hooks
Vireo-Telemetry (Backend)	`PayPal-Venmo/vireo-telemetry`	n8n + storage
Apple Virtualization CI/CD	`PayPal-Venmo/apple-virtualization-ci-cd`	Tooling + HLD
DX Sounds	`PayPal-Venmo/developer-experience-sounds`	MCP server (in AI Hub)
claude-cosmos	`PayPal-Venmo/claude-cosmos`	Internal Claude flavor
cosmos-code-mcp	`PayPal-Venmo/cosmos-code-mcp`	Fast Cosmos MCP server
Darwin	`gurisingh_paypal/darwin`	Evolution framework
ai-toolkit-registry-evolved	`gurisingh_paypal/ai-toolkit-registry-evolved`	Darwin staging (71 PRs)
devmo-beta-service contributions	`gurisingh_paypal/devmo-beta-service` PR #1 + #2	Prompt Enhancer + Guardrails
paypal-agent-zero	`gurisingh_paypal/paypal-agent-zero`	Agent-0 fork w/ soul
Cosmos-IDE	`gurisingh_paypal/Cosmos-IDE`	Local IDE wrapper
Hackathon-2025 / Mercury 2025	`github.paypal.com/gurisingh/Hackathon-2025`	Multi-agent merchant POC
video-toolkit *(new)*	`PayPal-Venmo/video-toolkit`	Remotion 4.0 + Veo/Chirp3 video pipeline
Darwin — org home + CI *(new)*	`PayPal-Venmo/Darwin` · `DPEIR-4171`	Productionized evolution framework
devmo (canonical)	`PayPal-Venmo/devmo` PR #59 + #60	Prompt Enhancer + Guard Rails (migrated from beta)

MAIA × Venmo (contributing track)

SURFACE	REFERENCE
Epic	`DTVZ-374`
P0 children	`DTVZ-375 … DTVZ-380` , `DTVZ-506`
Venmo P0 scoping (May 14)	`DTVZ-544 … DTVZ-555` (closed/accepted)
Execution tickets (Jun 10)	`DTVZ-670 … DTVZ-675`
MAIA repo	`paypalcorp/agenticmerch-ai-dev-maia`
Cross-functional channel	`#tmp-maia-venmo-xfn`
GTM brief	*Internal GTM Brief — Merchant AI Integration Agent (MAIA)* (BCGPS)
Executive deck	`MAIA_Venmo_Executive_Demo.pptx`
Demo recording	`Demo: AI Venmo Merchant Integrations - MAIA x Venmo (April 29 2026)`

Talks & artifacts

ITEM	REFERENCE
"How to Have the Most Fun with AI" (talk, Mar 6 2026)	Confluence VAC `2687045496` — slides on SharePoint `/p/gurisingh/IQDqxK9KbYcWT5ssEKlgZQU8…`
FunAI Hack Hour	Biweekly internal demo session (launched post-talk)
Soul Stack pattern	`SOUL.md` + `STYLE.md` + `SKILL.md` for agent persistence
OmniGraph / PWV Knowledge Graph presentation (Mar 31 2026)	`DTVZ-335` ; 38 nodes / 51 typed edges / 15 repos
Three-track framework memo	`#v-merchant-sandbox` Mar 31
DEVMO DOSSIER — Field Report (Apr 30 2026)	`#v-devmo`

Devmo Workstreams (owned)

WORKSTREAM	REFERENCE
Workstream 7 — Telemetry (Vireo)	Confluence VAC `2900859692`
Workstream 8 — Memory (Vireo memory store)	Same
Devmo AI tickets (Vireo + Prompt Enhancer)	`DTVAI-12` · `DTVAI-18` · `DTVAI-19` · `DTVAI-20` · `DTVAI-21`

Appendix B — Source receipts

Selected source links used to compose this portfolio (all internal):

- AI Topics — "How to Have the Most Fun with AI" (Mar 6 2026)
- Devmo Beta Service — Workstreams (Vireo workstreams 7 + 8)
- DTVZ-374 — Project MAIA Epic
- DTVZ-506 — MAIA Telemetry post-demo follow-on
- DTPLTM-8222 — Devmo-Telemetry (Vireo) Epic
- DTVZ-330 — cosmos-code-mcp
- DTVZ-331 — claude-cosmos
- DTVZ-335 — OmniGraph / PWV Knowledge Graph
- Confluence VAC `2860853837` — Draft HLD: Parallel iOS UI Test Execution Using Apple Virtualization.framework
- Slack — Mar 2 first share of Mac-in-a-Mac (22m 38s → 7m 23s)
- Slack — Mar 10 "Agent0 on the VM without docker"
- Slack — Mar 28 handoff prompt to a peer engineer
- Slack — Mar 31 Three-Track framework memo (`#v-merchant-sandbox`)
- Slack — Apr 9 to the Livery lead: Livery × Mac-in-a-Mac × MAIA
- Slack — Apr 14 internal update "332 → 9,513 in One Weekend"
- Slack — Apr 14 Vireo ownership announcement in `#v-devmo`
- Slack — Apr 27 MAIA × Venmo Sync Agenda + Script to my manager
- Slack — Apr 28 MAIA Singapore call recap (a leadership group DM)
- Slack — Apr 30 DEVMO DOSSIER field report

- Slack — May 1 Prompt Enhancer PR #1 + guardrails PR #2
- Slack — May 4 DX Sounds merge announcement
- Slack — May 5 Mac-in-a-Mac pitch to an iOS leadership sponsor
- Slack — May 8 Darwin thesis to the AI Toolkit lead
- Email — Apr 30 the MAIA PM's "MAIA X Venmo Demo: Recap & Next Steps"
- Email — May 4 the program lead unifies the program
- **Mid-year (May–June 2026) receipts:**
- Slack — May 12 `video-toolkit` announced in `#pwv-bazaar`
- Slack — May 15 Darwin "home stretch" update to the AI Toolkit lead
- Slack — May 15 MAIA telemetry recap (Vireo v2.1: 53 / 45 / 40 / 10, 100%)
- Slack — May 28 Devmo DTVAI tickets in `#v-devmo`
- Slack — Jun 2 Hermes security accounting (`#ae_cir27_2954854_hermes-ai-agent`)
- Slack — Jun 8 the MAIA PM names the MAIA "build" core
- Slack — Jun 23 "Sutter Pay With Venmo + MAIA" demo
- Slack — Jun 24 the MAIA PM: "plug Venmo into the one MAIA roadmap you own"
- Jira — `DTVZ-544…555` (Venmo P0) · `DTVZ-670…675` (execution) · `DPEIR-4171` (Darwin CI) · `DTVAI-18…21` (Devmo)
- Email — Jun 24 "MAIA × Venmo: XFN Working Group" (the MAIA PM), canceled Jun 26 (*"combining with other efforts"*); Jun 17 "MAIA Cohorting"; Jun 4 "Devmo Syncs" (AWS/Stripe benchmarking)

This document is for internal calibration and the mid-year check-in. Last updated 2026-06-26.